

ML-KEM CBD RTL optimization on VTR

Post-quantum cryptographic IP / open 45 nm FPGA proxy

A fixed-chunk state representation reduces the paired area-delay-power estimate by 9.734% across 64 prospectively frozen publication pairs.

Source: HOPE-MLKEM CBD at pinned commit [72a90d80](#) . . .

Evidence completed: 21 August 2026

Academic comparative estimate. Not ASIC signoff, commercial-FPGA characterization, side-channel analysis, or certification of the complete ML-KEM implementation.

The result a technical customer can review first

1. Executive result

The optimized RTL preserves the complete interface and defined-input cycle behaviour while improving the fixed 64-pair composite estimate by 9.734% (two-sided 95% descriptive interval: 9.333% to 10.133%).

Metric	Baseline	Optimized	Improvement	95% interval
Total area (MWTA)	22,751,867	19,132,428	15.908%	15.526% to 16.288%
Critical path	4.0538 ns	3.7703 ns	6.992%	5.472% to 8.487%
Active total power	64.106 mW	60.284 mW	5.962%	4.034% to 7.852%
Composite score	1.000000	0.902662	9.734%	9.333% to 10.133%

MWTA is VTR's minimum-width transistor-area unit. The optimized implementation uses 270 packed CLBs instead of 337 (19.881% fewer). Reciprocal routed delay corresponds to a 7.518% increase in the frequency estimate. The score is the equal-weight geometric mean of paired area, delay and active-total-power ratios; percentages are not added.

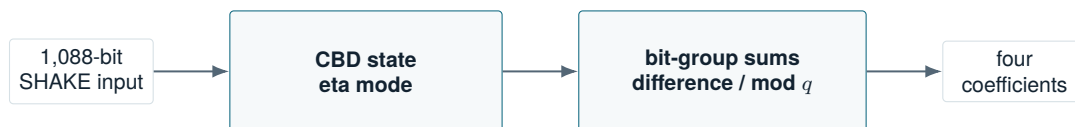
Publication design. The RTL was frozen before a new pool of 64 seeds was measured. A prior fixed 12-pair result had shown 9.465% improvement and is disclosed as selection history. The publication pool was not extended or edited after measurement.

A real post-quantum cryptographic building block

2. Module and industrial context

ML-KEM is the module-lattice key-encapsulation mechanism standardised in NIST FIPS 203. The evaluated module is the centred-binomial-distribution (CBD) sampler from the open [HOPE-MLKEM](#) hardware implementation. It converts pseudorandom SHAKE bits into small positive and negative polynomial coefficients used as controlled mathematical noise.

This is one block inside ML-KEM, not a complete key-encapsulation engine. Its relevance is nevertheless concrete: it is synthesizable RTL from a real post-quantum design, with the state movement, modes and command behaviour expected in cryptographic IP.

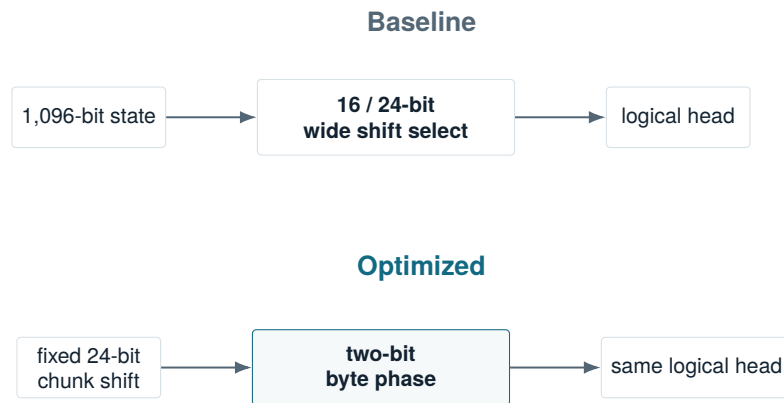


The source is pinned to commit `72a90d80484d45d0bed1e0f9903bd0fb78cceb47`, path `rtl/cbd.v`, under the MIT License. The evaluation baseline differs from upstream only by provenance comments.

Replace wide variable movement with fixed movement plus metadata

3. Architectural change

The baseline stores 1,096 bits in `reg_shake`. Each active cycle selects either a 16-bit or a 24-bit right shift according to `eta`. Generic mapping exposes this as a wide selection network around more than one thousand enabled flip-flops.



The optimized RTL always moves physical chunks by 24 bits. A two-bit phase records the remaining byte offset needed to represent a logical 16-bit advance. This is analogous to retaining data in place and moving a read pointer instead of repeatedly moving the full data structure.

The public interface, addresses, output coefficients, completion cycle, reset, load, second-load, stalls and both `eta` modes remain unchanged.

The exact source change is published as `rtl/changes.patch`.

Sequential behaviour, not only arithmetic equivalence

4. Correctness contract

The candidate passes three layers before any PPA metric is accepted:

Gate	Scope	Result
Structure and lint	Verilator lint; Yosys module, interface, hierarchy and synthesis checks	PASS
Cycle regression	239 cycles and 239 checks covering reset, eta 2/3, stalls, reload and second-load	PASS
Formal equivalence	EQY SAT five-step temporal induction after two reset cycles; complete joined 1,096-bit logical-state refinement	PASS

Five-step induction is not a five-cycle bounded claim. The induction establishes an invariant for arbitrary future cycles after reset. Public outputs remain final obligations, and arithmetic temporaries are excluded as accidental match points.

Formal boundary. Equivalence uses two-state, defined-input hardware semantics after the declared reset. It does not claim identical Verilog X/Z propagation. The full archive contains the configuration, testbench and raw pass logs bound to the optimized RTL hash.

Baseline RTL SHA-256:

c152f96a682ea390c610dc32ee00a317c0ad990f9b082451325cdaf12f2218b0

Optimized RTL SHA-256:

da0b4e79beb9cf9cc24c3ce949068c3a29316273a6320e283f4201fbd884c9ef

What may not change after candidate selection

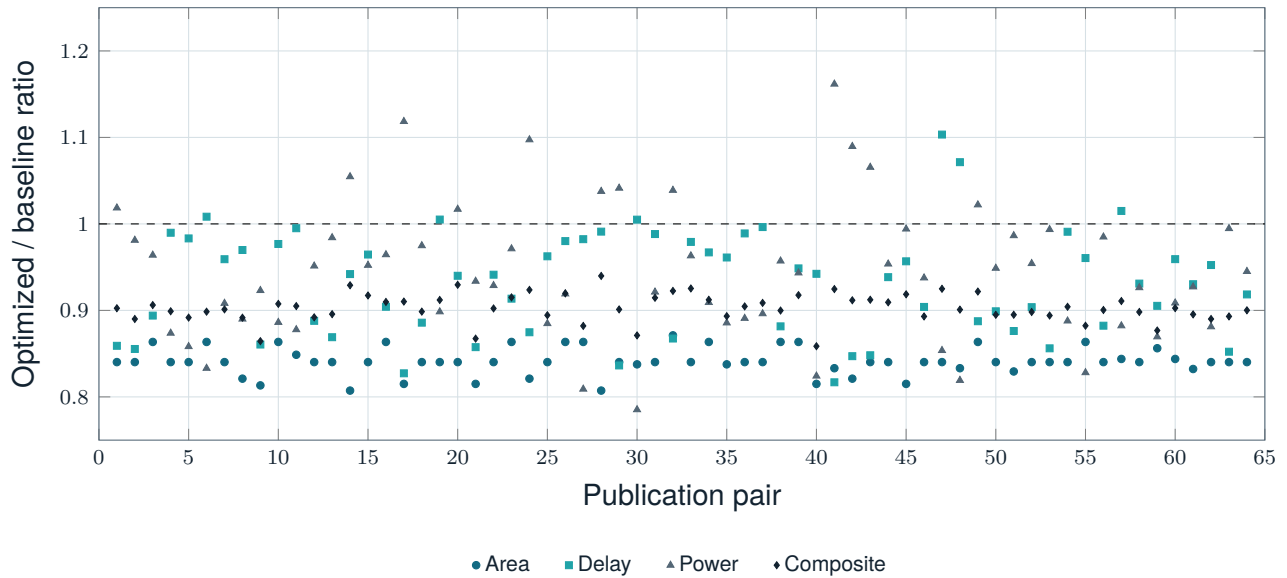
5. Frozen evaluation contract

Dimension	Frozen choice
Editable artifact	Optimized <code>cbd.v</code> only; final hash fixed before publication measurements
Functional behaviour	Exact ports, reset, loading, modes, stalls, addresses, termination and cycle outputs
Physical flow	Pinned VTR Linux/amd64 image, k6-N10 LUT6 architecture and PTM 45 nm model
Runtime envelope	Network disabled, 2 CPU, 7 GiB RAM per container
Power activity	ACE deterministic probabilities: input static 0.5, input switching 0.2, clock 0.2, seed 1
Search	Seeds 1, 7 and 19; optimizer-visible only
Initial confirmation	Twelve fixed disjoint pairs; observed before publication protocol
Publication confirmation	Seeds 101 through 164; fixed $n = 64$ with no extension
Primary score	Equal-weight paired geometric mean of area, delay and active total power
Acceptance	Composite one-sided 95% upper log bound below zero; no primary metric wholly regressed

Changing the target, activity model, seed pool, score or correctness semantics produces a different experiment. It cannot be represented as an improvement within this result.

Variation across the 64 frozen implementation pairs

6. Paired evidence



Area and CLB count reflect the mapped structure plus routing. Delay and power vary with placement seed, which is why acceptance uses paired ratios rather than comparing one favourable implementation. The confidence intervals quantify implementation-seed variability for this exact academic target; they do not represent semiconductor process variation.

A small review path with a deeper audit path

7. Evidence package

The Git repository contains the before/after RTL, exact patch, compact certificate, report, checksums and a fail-closed Python verifier. A customer can check the complete compact package with:

```
python3 verify.py
```

The separately downloadable raw archive contains all baseline and optimized timing and ACE power outputs for the 64 pairs, correctness logs, formal inputs, publication protocol and an internal hash manifest. Supplying it to the same verifier additionally re-extracts every primary metric:

```
python3 verify.py --evidence-archive \  
  mlkem-cbd-vtr45-full-evidence-v1.tar.gz
```

The compact verifier rejects altered RTL, patch drift, metric or score substitution, non-finite JSON, authority changes and incomplete manifests. Full-evidence mode also rejects unsafe archive paths, unexpected files, host paths, credential markers, failed routes and any raw metric that differs from the certificate.

Audit meaning. Raw-provenance verification checks the recorded files and re-derives the result; it does not rerun the EDA tools. A fresh external reproduction remains a separate, stronger review step.

Certificate SHA-256:

```
3d8b6120db2957da6c4d3cd4569e73df180ba07d5c624c9ae4bc1f4d2d37406d
```


Why this case complements SHA-1 and INT8 MatVec

8. Portfolio and customer relevance

The three public cases demonstrate different optimization classes:

Case	Domain	Transformation class
SHA-1	Legacy sequential cryptography	Shared Boolean structure and direct state readout
INT8 MatVec	Quantized AI arithmetic	Range-correct widths and balanced accumulation
ML-KEM CBD	Post-quantum cryptographic IP	Alternative physical state representation with a formal refinement relation

ML-KEM CBD is particularly relevant to teams implementing cryptographic FPGA/ASIC IP or preparing post-quantum migration. The case demonstrates an engagement pattern rather than a product claim: freeze one owned block and target, prove exact behaviour, measure an implementation objective and deliver auditable before/after evidence.

Cross-case percentages are not an apples-to-apples benchmark. Each circuit has its own interface, workload or activity model and implementation distribution. The supported claim is the paired improvement inside each frozen case.

What the evidence does not establish

9. Limitations

This result does not claim:

- ASIC synthesis, place-and-route or signoff;
- Vivado, Quartus or commercial-FPGA device performance;
- physical-board, measured-power, measured-energy or manufactured-silicon results;
- side-channel resistance, masking quality, constant-time software behaviour or fault resistance;
- security certification of HOPE-MLKEM or a complete FIPS 203 implementation;
- application-workload power, because ACE probabilistic activity is used;
- that the same state representation improves a different technology, architecture or integration context;
- accredited certification or independent third-party assurance.

The open PTM 45 nm FPGA proxy is useful for controlled relative experimentation. A customer pilot should replace it with the customer's implementation target, activity, constraints and signoff boundary.

Conclusion. Within the exact published VTR/PTM45 contract, the frozen optimized CBD RTL passes sequential correctness gates and reduces the prospectively confirmed paired PPA estimator by 9.734%. The result is understandable from the compact package and auditable down to every raw publication-pair metric.